

TrES-1b

R-2386

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120623 · created 2026-08-20T23:12:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456101.73618 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.082 +/- 0.025
Transit depth (Rp/Rs)^2	0.67 +/- 0.42 [%]
Orbital Inclination (inc)	88.98 +/- 1.22
Airmass coefficient 1 (a1)	1.021 +/- 0.012
Airmass coefficient 2 (a2)	-0.0215 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#3 - [259, 179]
Optimal Aperture	3.36
Optimal Annulus	12.35
Transit Duration (day)	0.0955 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.082 ± 0.025 vs 0.1358 (deviation 1.45σ)
Duration fit vs published	0.0955 d vs 0.1042 d (deviation 1.02σ)
Mid-time O–C	2.2 min
Depth SNR	2.2
Residual scatter	1.19 %

Light curve

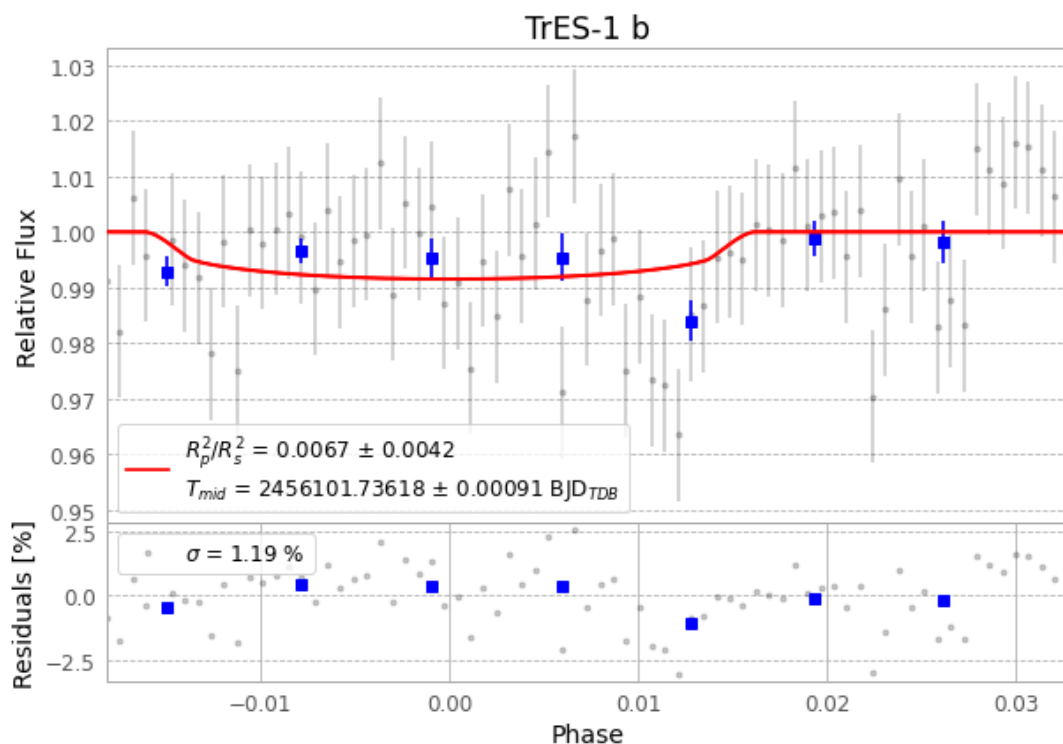


Figure 1 — FinalLightCurve_TrES-1 b_2012-06-22.png (SHA-256 478d108767097c0a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 74f567c9c222c7dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), TRES-1120623041512.FITS → TRES-1120623075713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120623.log (36385ac9ccca...), FinalLightCurve_TrES-1 b_2012-06-22.png (478d10876709...), FinalLightCurve_TrES-1 b_2012-06-22.csv (3b0a80e8c2e3...)

Compute resources

Wall time 117.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:12Z.